

THE UNIVERSITY OF HONG KONG

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE

Final Examination

COMP3314 Machine Learning

18 Dec 2023, 2:30 pm - 4:30 pm

- Candidates are permitted to bring to the examination ONE sheet(s) of A4-sized paper with printed/written notes on both sides.
- There are 6 questions to be answered. Answer all the questions (total: 100%) in the spaces provided.
- For problems that involve derivations, credit will be given only if the work leading to the answer is shown.
- Write your University No. on every page. Do not write your name on this examination script.
- Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.
- Brand and Model of the calculator: _____

Q1	20 points	
Q2	16 points	
Q3	12 points	
Q4	16 points	
Q5	20 points	
Q6	16 points	
Total	100 points	

Problem 1

TRUE or FALSE questions.

1. (2 points) For three random variables, X, Y, Z . If X, Y , and Z are pairwise independent, then the three variables are mutual independent, i.e., $p(X, Y, Z) = p(X)p(Y)p(Z)$.
2. (2 points) Given a 2-dimensional dataset with N points $(x_i, y_i)_{i=1}^N$, the label of each point is either positive or negative. There exists a neural network $f_\theta(x, y) = w_1x + w_2y + b$, where $\theta = [w_1, w_2, b]$, $w_1, w_2, b \in \mathbb{R}$, that can separate all positive and negative points.
3. (2 points) Bayes' rule could be written as $\text{likelihood} \propto \text{posterior} \times \text{prior}$
4. (2 points) You are training a neural network on the ImageNet dataset, and you are thinking of using gradient descent as your optimization function. It is not possible for Adam optimizer to converge faster than Stochastic Gradient Descent optimizer.
5. (2 points) L1 regularization leads to weight sparsity, while L2 regularization leads to smaller weights.
6. (2 points) After training a neural network, you observe a large gap between the training accuracy (100%) and the test accuracy (42%). You can decrease the batch size to reduce the gap.
7. (2 points) During backpropagation, as the gradient flows backward through a sigmoid non-linearity, the gradient will always decrease in magnitude.
8. (2 points) Generative models can be used for classification.
9. (2 points) In KNN implementation, You can't save memory of stored points without compromising performance.
10. (2 points) If k-means clustering is run long enough, it will always terminate.

Answer 1

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

Problem 2

Multiple Choice Questions (one or more choices are correct)

1. (4 points) Let us assume we implement an AND function to a single neuron. Below is a tabular representation of an AND function:

Table 1: The AND dataset consisting of four points.

ID	x_1	x_2	y
a	0	0	0
b	0	1	0
c	1	0	0
d	1	1	1

The activation function of our neuron is denoted as:

$$f(x) = \begin{cases} 0, & \text{for } x < 0 \\ 1, & \text{for } x \geq 0 \end{cases}$$

Now we express the neuron as $y = f(w_1x_1 + w_2x_2 + b)$ and use this single-neuron to classify the dataset. What would be the weights and bias?

- A. $b = -1.5, w_1 = 1, w_2 = 1$

B. $b = 1.5, w_1 = 2, w_2 = 2$

C. $b = 1, w_1 = 1.5, w_2 = 1.5$

D. None of the above
2. (4 points) Which of the following would you consider to be valid activation functions (non-linearities) to train a neural net in practice?
- A. $f(x) = -\min(2, x)$

B. $f(x) = 0.9x + 1$

C. $f(x) = \begin{cases} \min(x, .1x) & | x \geq 0 \\ \min(x, .1x) & | x < 0 \end{cases}$

D. $f(x) = \begin{cases} \max(x, .1x) & | x \geq 0 \\ \min(x, .1x) & | x < 0 \end{cases}$
3. (4 points) Suppose we are given data comprising points of several different classes. Each class has a different probability distribution from which the sample points are drawn. We do not have the class labels. We use k-means clustering to try to guess the classes. Which of the following circumstances would undermine its effectiveness?
- A. Some of the classes are not normally distributed.

B. The variance of each distribution is small in all directions.

C. Each class has the same mean.

D. You choose $k = n$, the number of sample points.
4. (4 points) Assume the vectors v and u are defined as numpy arrays. The vectors are multiplied together using the $*$ operator: $v*u$. Which of the following shapes are valid, meaning the operation will succeed without errors?
- A. $v = (4, 1), u = (4, 1)$

B. $v = (1, 4), u = (50, 4)$

C. $v = (1, 1), u = (50, 4)$

D. $v = (1, 4), u = (50, 1)$

Answer 2

- 1.
- 2.
- 3.
- 4.

Problem 3

(*Naive Bayes Classifier*)

In this question, you are given the following set of data and try to solve a series of problems. You need to determine how three different features contribute to whether an advertisement is clicked by users. In this table, there are 8 data points representing typical users.

Table 2: The recorded 8 data points.

Advertisement	Animated?	Popup?	Colorful?	Clicked
1	Yes	Yes	No	No
2	No	Yes	Yes	No
3	No	Yes	No	No
4	No	No	Yes	Yes
5	Yes	No	Yes	Yes
6	Yes	No	No	Yes
7	No	No	Yes	Yes
8	No	No	No	No

1. (4 points) According to the naive Bayes classifier, what is a user clicks a popup ads $P(\text{Clicked} = 1 | \text{Popup} = 1)$?
2. (4 points) According to the naive Bayes classifier, what is the probability that a clicked ads is colorful and animated, i.e. $P(\text{Animated} = 1, \text{Colorful} = 1 | \text{Clicked} = 1)$?
3. (4 points) How many parameters are there in the naive bayes classifier $P(\text{Clicked} | \text{Popup})$?

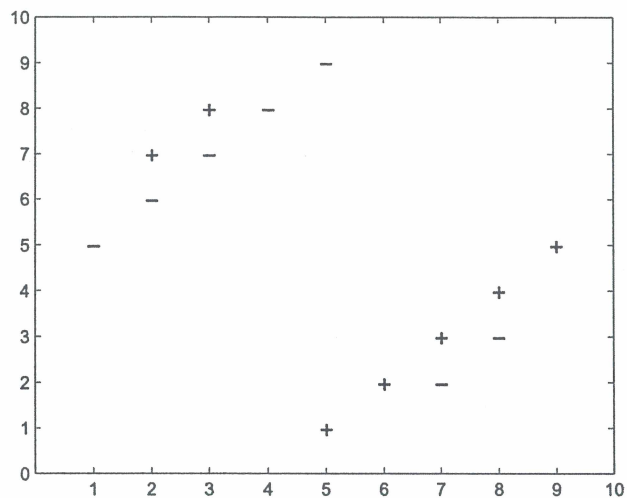
Answer 3

- 1.
- 2.
- 3.

Problem 4

(*k*-nearest neighbor) In the following questions you will consider a *k*-nearest neighbor classifier using Euclidean distance metric on a binary classification task. We assign the class of the test point to be the class of the majority of the *k* nearest neighbors. Note that a point can be its own neighbor.

- (6 points) What value of *k* minimizes the training set error for this dataset? What is the resulting training error?
- (6 points) Why might using too large values *k* be bad in this dataset? Why might too small values of *k* also be bad?
- (4 points) Sketch the 1-nearest neighbor decision boundary for this dataset.



Answer 4

1.

2.

3. Plot in the figure above

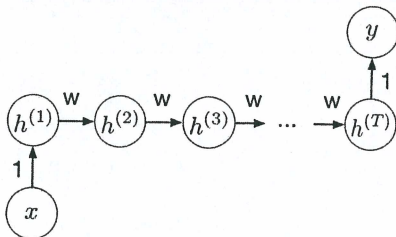
Problem 5 (Exploding/Vanishing Gradients of RNN)

Consider the following RNN, which has a scalar input at the first time step, makes a scalar prediction at the last time step, and uses a shifted logistic activation function:

$$\begin{aligned} z_t &= W * h_t \\ h_{t+1} &= \phi(z_t), t < T \\ h_1 &= x \\ y &= h_T \\ \phi(z) &= \sigma(z) - 0.5 \end{aligned}$$

[Note: All expressions should be in terms of $\frac{\partial \mathcal{L}(\hat{y}, y)}{\partial \hat{y}}$, $\sigma'(x)$ and variables specified above, e.g input into activation function at time t , z_t .]

1. (4 points) Compute the gradient of $L(\hat{y}, y)$ with respect to h_{T-1} using the back-propagation algorithm.
2. (4 points) Let $\bar{h}_t$ be the gradient of h_t to weights, i.e., $\bar{h}_t = \frac{\partial h_t}{\partial W}$, express $\bar{h}_t$ as a function of $\bar{h}_t$, when $t < T$, using the back-propagation algorithm.
3. (6 points) Explain with the expression of the gradient $\frac{\partial \mathcal{L}(\hat{y}, y)}{\partial W}$, why the gradients of RNN often explodes $\rightarrow \infty$ or vanishes $\rightarrow 0$ during training. [Hint, you will be using the chain rule, i.e. when z is dependent on y , which itself depends on x , $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x}$]
4. (6 points) Suppose the input to the network is $x = 0$. Notice that $h_t = 0$ for all t . Determine the value α such that if $W < \alpha$, the gradient vanishes, while if $W > \alpha$, the gradient explodes. You may use the fact that $\sigma'(0) = 1/4$.


Answer 5

1.

2.

3.

4.

Problem 6

(Gradient Descent for k -means Clustering)

K-means is a popular method to cluster (or partition) a set of N data points into K clusters, so that each data point belongs to the cluster with the nearest cluster center (or called **centroids**).

Technically, let us denote the K clusters as $\mu_1, \dots, \mu_K \in \mathbb{R}$ and the set of N data points as $x_1, x_2, \dots, x_N \in \mathbb{R}$. To cluster the data such that each data point is assigned to its closest centroid, the algorithm proceeds as follows:

- (1) Randomly initialize the cluster centroids μ_k for each $k = 1, 2, \dots, K$;
- (2) For each data point x_n with $n = 1, 2, \dots, N$, compute the index z_n of its closest centroid,

$$z_n = \arg \min_k \|x_n - \mu_k\|_2^2;$$

- (3) Given the information contained in data points, we compute the new cluster centroids as

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n: z_n = k} x_n,$$

where N_k is the number of data points that are assigned to the k -th cluster.

- (4) Compare the old and new cluster centroids and if their difference is small enough (say less than $1e-8$), then the algorithm has converged and terminates; otherwise, return to step **2** and iterate over the entire data set. In both cases, update the centroids as

$$\mu_k \leftarrow \mu_k^{\text{new}}, \quad k = 1, 2, \dots, K.$$

An important metric to reflect the clustering quality is the **distortion**, which measures how individual data points deviate from their assigned cluster centroids. It is just the mean of the squared distance between each data point and its assigned centroid,

$$J = \frac{1}{N} \sum_{n=1}^N \|x_n - \mu_{z_n}\|^2.$$

For now, suppose we have computed the cluster assignments z_n for each data point in Step 2 and we would like to update the cluster centroids.

1. (4 points) Show that the update rule in Step 3 & 4 is equivalent to minimizing the distortion J . [Hint: break the sum $\sum_{n=1}^N$ into $\sum_{k=1}^K \sum_{n: z_n = k}$.]
2. (6 points) We now consider a modified K-means algorithm, where instead of employing the centroid update rule in Step 3 & 4, we update centroids by directly minimizing the distortion with stochastic gradient descent (SGD). Derive the gradient descent rule for each centroid $\mu_1, \dots, \mu_K$ on a single data point x_i , with a per-cluster learning rate α_k . [Hint: the data point x_i can only be assigned to one cluster with index z_i .]
3. (6 points) Assume we use an SGD strategy where we simply iterate all data points from x_1 to x_N (each data point will be visited once and only once). Compute the learning rate α_k for each cluster so that the resulting SGD update will be equivalent to the update rule in Step 3 & 4 after it has iterated over all data points.

Answer 6

1.

2.

3.

-END OF PAPER-